

Analysis Model and Data

Modeling

The driving question of this project is to understand how two different grocery store subsidy policies impact the welfare of consumers. These policies are a blanket subsidy, given as a rent subsidy or rent break, and a contracted 30% off of core basket items, as defined by the NYC Groceries initiative. We propose a simplified model in order to capture the main drivers of grocery behavior, making reasonable assumptions to allow us to capture the main effects of this policy. In this static analysis, the consumers will change their preferred grocery store after the government subsidizes a store, but for the purposes of our analysis, the surrounding firms do not make any strategic response. Therefore, we need to model the consumers, the firms, consumer choices that incorporate prices (at least), the amount firms lower prices, and a measure of welfare. The ultimate decision goal is to find the lowest subsidy to achieve a certain welfare gain, and compare that subsidy number across stores.

To begin, we model consumers by census tract t , as it is the most granular division with usable data. We further divide households in each tract into three income groups g , since deciding factors for shopping depend on income and the amount spent per household also depends on income. The set of stores J includes grocery stores that offer a full set of groceries (produce, meat, dairy, non-perishables) that would be eligible for the NYC Groceries contract. In order to assign a benefit to each consumer group for each store in the set, we use the economic concept of “utility,” which does not have interpretable units, but has the ordinal property of being able to rank stores by the consumer’s preferences. The significant drivers of supermarket choice are the size (measured in square footage) sq_j , the difference between the average price of a standard basket of goods for store j p_j and the average standard basket price for all stores $\bar{p}$, and distance to the store d_{ij} . The price term is multiplied by the number of baskets the income group purchases per trip, which is calculated as $b_i = 52\bar{f}_g / (T \cdot \bar{p})$ where T is the number of trips taken per year. Note that when $T = 52$, the number of baskets is simply the average yearly spending on food at home $\bar{f}_g$ divided by the average price of the standard basket $\bar{p}$. Lastly, to differentiate between the added variety of the larger chain stores and the smaller independent stores, a type term α_j is included, giving the final systematic utility equation

$$V_{ij} = \alpha_j + \gamma_{sq}(sq_j/1000) - \beta_{p,i} b_i(p_j - \bar{p}) - \beta_d d_{ij} \quad (1)$$

There are other unobserved factors when consumers are choosing stores, so we model it with an error term ϵ_{ij} , and since the consumer is maximizing utility, it follows a Gumbel distribution (Type I extreme value) (CITE SOURCE).

$$U_{ij} = V_{ij} + \epsilon_{ij}, \quad \epsilon_{ij} \stackrel{\text{i.i.d.}}{\sim} \text{Gumbel}(0, 1)$$

Now that we have the utility of each consumer group for each store, we construct a method for estimating the probability that the consumer group will attend that store using the utility. The logit of a probability is the log-odds of the consumer making a choice.

$$\text{odds}(p) = \frac{p}{1-p}, \text{logit}(p) = \ln\left(\frac{p}{1-p}\right)$$

Since the consumer can make more than two choices, we use a multinomial logit function. Following (CITE SOURCE), when integrating over ϵ_{ij} , we get a closed form solution for the probability s_{ij} that consumer group i chooses store j over all other stores

$$s_{ij} = \Pr(U_{ij} \geq U_{ik} \quad \forall k) = \frac{\exp(V_{ij})}{\sum_{k=1}^J \exp(V_{ik})}$$

Since consumers are also purchasing food at smaller store types as well as stores in other community districts, they are modeled simply as “other stores” $j = 0$ where indirect utility is set

at a reference of $V_{i0} = 0$, thus leaving the sum of all shares $\sum_j s_{ij} < 1$ where the leftover amount

$1 - \sum_j s_{ij}$ is the share of consumers group i spending at “other stores.” The logit choice

probability equation can be rewritten as

$$s_{ij} = \frac{\exp(V_{ij})}{1 + \sum_{k=1}^J \exp(V_{ik})}$$

It follows then that the expected utility is the log sum of the _____

$$E[\max_k U_{ik}] = \ln \left[\sum_{k=1}^J \exp(V_{ik}) \right] + \gamma$$

where γ is the Gumbel average, and the annual welfare is found by multiplying by the number of trips and dividing by the price sensitivity parameter $\beta_{p,i}$ because _____. To find the change in consumer surplus, subtract the welfare before the change from after the change, multiply by the number of households in the consumer group n_i and sum for all groups

$$\Delta CS = \sum_i \frac{n_i \cdot T}{\beta_{p,i}} \left[\ln \sum_k \exp(V_{ik}^{\text{post}}) - \ln \sum_k \exp(V_{ik}^{\text{pre}}) \right]$$

The final step is modeling the change in price for a store that receives the subsidy Δp_j . The simple case is when a rent payment or a tax break is given to a store, where the market leads to a “pass-through” rate θ , dependent on the size of the subsidy. The change in basket price for the store is

$$\Delta p_j = \theta \cdot \frac{s}{\bar{Q}_j}$$

where $\hat{Q}_j$ is the predicted number of baskets sold per year before the policy. (how does this equation make sense?)

The second case is when the store is contracted to provide “core basket” items at a 30% discount, and offered full coverage of rent and property taxes with the option of an additional “affordability payment” AP to cover the cost. This reduction of prices depends on the store basket price p_j and the share of total spending spent on core basket items for each income group κ_g .

$$\Delta p_{j,g} = 0.30 \cdot \kappa_g \cdot p_j$$

Then, the share of core basket sales at store j is the sum of the total amount spent by all income groups on core goods divided by the amount spent by all income groups on all goods. (why is there no price term here?)

$$\kappa_j = \frac{\sum_i n_i \cdot s_{ij}^{\text{post}} \cdot b_i \cdot \kappa_g}{\sum_i n_i \cdot s_{ij}^{\text{post}} \cdot b_i}$$

This is used to find the required subsidy amount, which will cover 30% of core good sales, where $\hat{R}_j^{\text{post}}$ is the estimated revenue after the price drop and reallocation of consumers.

$$s_j^* = \text{Rent}_j + \text{Tax}_j + AP_j = \max(\text{Rent}_j + \text{Tax}_j, 0.30 \cdot \kappa_j \cdot \hat{R}_j^{\text{post}})$$

The objective of the model is to find the lowest subsidy to achieve a defined change in consumer surplus $\Delta CS_{\min}$ (a decision variable)

$$s_j^* = \min_{s_j} s_j \quad \text{subject to} \quad \Delta CS_j(s) \geq \Delta CS_{\min}$$

Data

- Consumer data
- Business data
- Economic parameters

Model Calibration

- Alpha & beta
- Note: when solving for when stores are given the subsidy directly, the betas shouldn't be calibrated to the NYC store because it doesn't exist. Instead, it should calibrate to the newly lowered price store.
 - There are clear issues with this. Many stores are located within the same _____ radius, and so it's likely more customers are willing to go there. It would clearly mess with competition in the area and put enormous pressure on the other stores